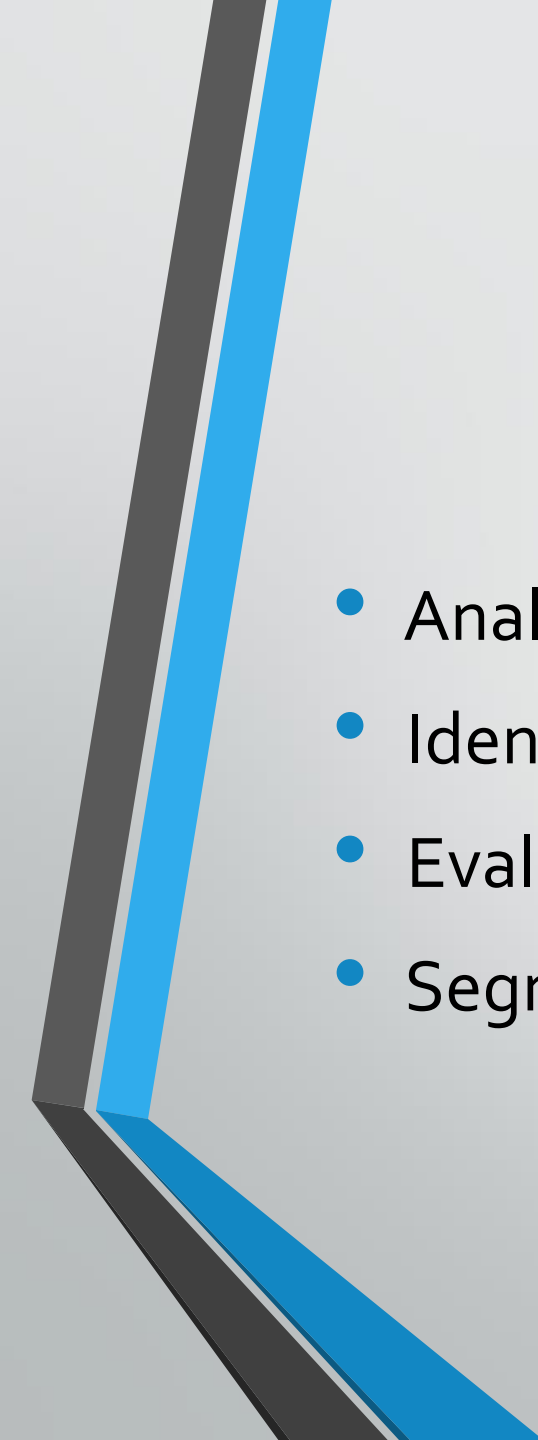




Customer Segmentation & Revenue Analysis

Tools: Python, K-Means Clustering, Power BI

Author: Vikram Bhagnani



Business Problem

Objective of the Project

- Analyze customer purchasing behavior using retail transaction data
- Identify product categories where discounts should be applied
- Evaluate spending patterns across age, season, and location
- Segment customers to enable targeted marketing strategies

Dataset Overview

- Dataset contains 3900 customer transactions
- Key variables: Age, Category, Purchase Amount, Review Rating
- Includes Season, Location, Subscription Status, Payment Method
- Data used to understand purchasing patterns and customer behavior



Data Preparation

Data Cleaning & Feature Engineering

- Checked for missing values and duplicates
- Created **Age Group feature** for demographic analysis
- Aggregated customer-level data for clustering
- Standardized variables before applying clustering

Exploratory Data Analysis (EDA)

- Revenue distribution analyzed across product categories
- Spending behavior evaluated across age groups
- Seasonal trends and location impact analyzed
- Identified major contributors to overall revenue

Revenue Insights Dashboard

Customer Revenue & Performance Overview

Season



3900

Total Customers

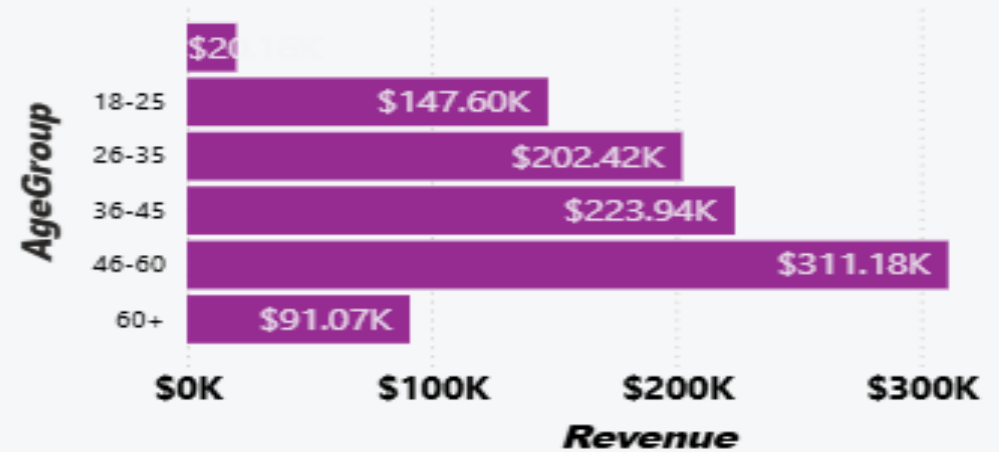
\$996.38K

Total Revenue

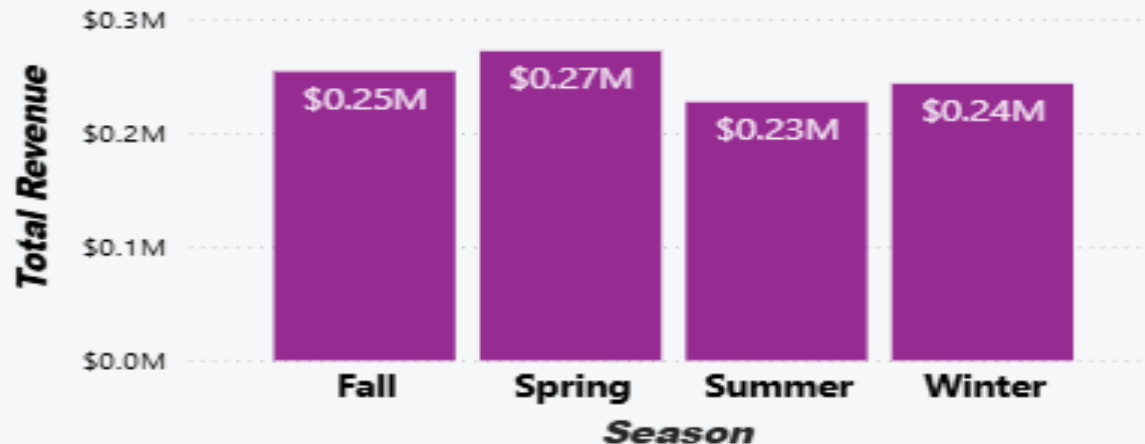
255.48

Avg Purchase Value

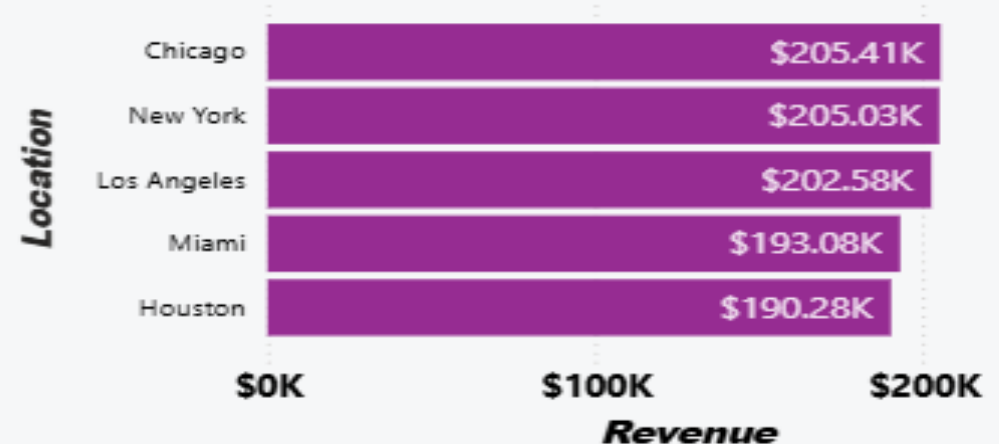
Revenue by AgeGroup



Total Revenue by Season



Revenue by Location



- **Insights**

- Customers aged 46–60 generate the highest revenue
- Chicago and New York Location contribute the most sales
- Spring season shows the highest revenue

Discount Strategy Recommendation

Category	Avg Spend Without Discount	Avg Spend With Discount	Insight
Clothing	254	270	Positive response
Accessories	255	252	Minimal impact
Electronics	252	249	Weak response
Home Decor	265	253	Weak response
Sports	252	250	Weak response

➤ Where Discounts Should Be Applied

- Based on category analysis:
 - **Clothing category responds strongly to discounts**
 - Accessories and Home Decor already perform well without heavy discounts
 - Recommendation:
- Apply **targeted discounts to Clothing** to increase conversion and basket size.

Customer Segmentation Approach

Clustering Methodology

- Customer-level dataset created by aggregating transaction data
- Features standardized before clustering
- K-Means clustering applied to identify customer segments
- Optimal number of clusters selected using Elbow Method

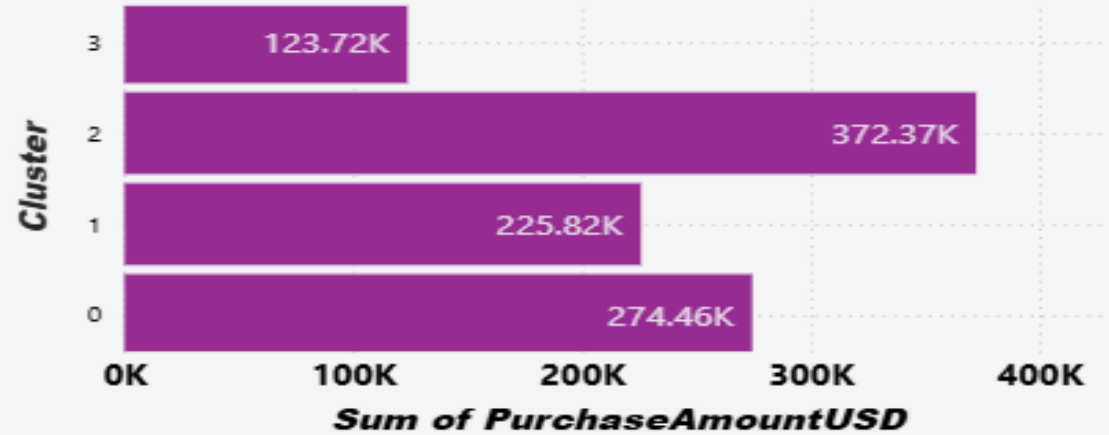
Customer Segmentation Dashboard

Customer Segmentation Insights

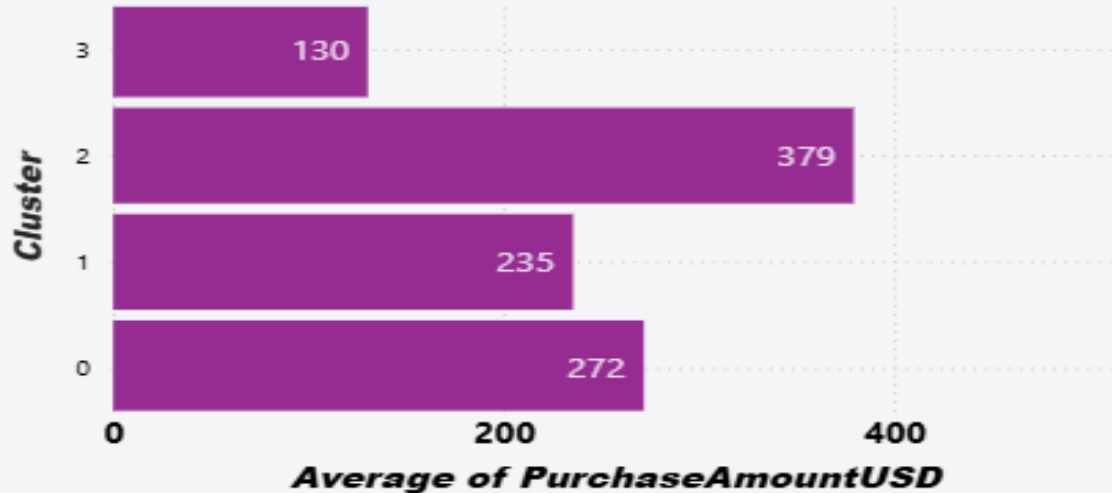
Customers per Cluster



Revenue by Cluster



Avg Spend per Cluster



Subscription Rate



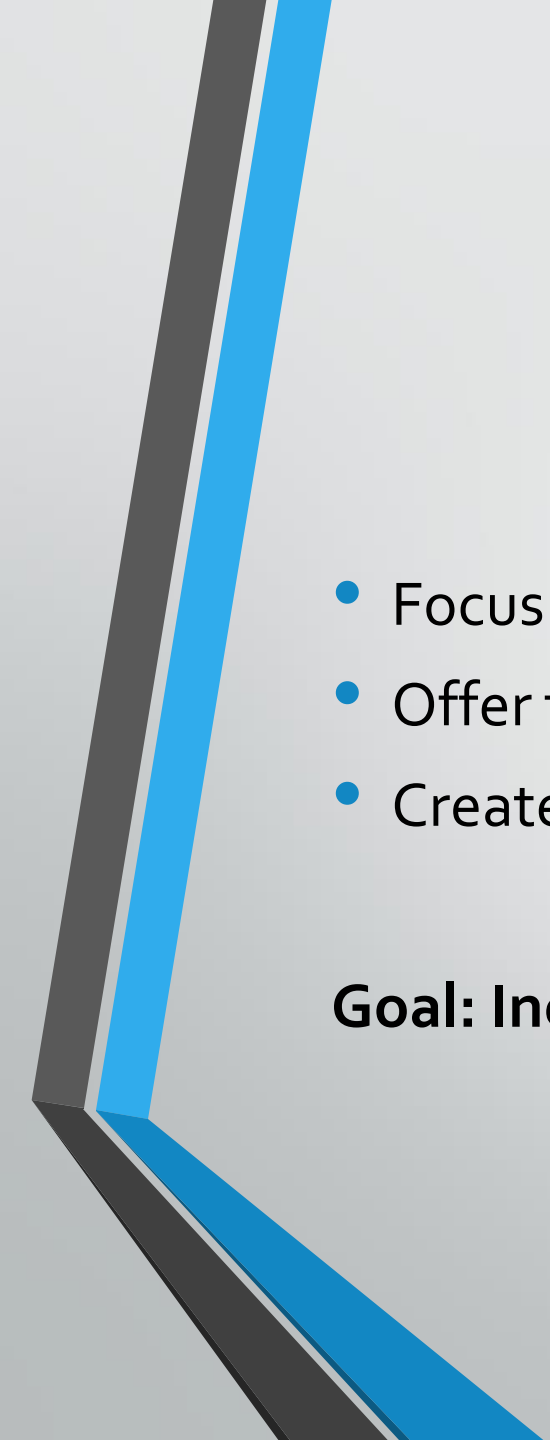
- **Insights from clustering:**

- Cluster 2 customers show the **highest average purchase value**
- Cluster 3 customers contribute the **lowest revenue**



Cluster Profiles

Cluster	Description
Cluster 2	High-value customers
Cluster 0	Regular customers
Cluster 1	Moderate spenders
Cluster 3	Low-value customers



Marketing Strategy

Recommended Business Actions

- Focus loyalty programs on **high-value customers**
- Offer targeted promotions to **moderate spenders**
- Create engagement campaigns for **low-value customers**

Goal: Increase customer lifetime value.



Business Impact

Expected Business Benefits

- Improved customer targeting
- Better marketing ROI
- Increased customer retention
- Data-driven decision making

Conclusion

Project Summary

- Identified key revenue drivers
- Segmented customers using machine learning
- Developed actionable marketing strategies

This approach enables businesses to **increase revenue and improve customer engagement.**



THANK YOU